



Improving End-to-end Mixed-case ASR with Knowledge Distillation and Integration of Voice Activity Cues

Sashi Novitasari, Takashi Fukuda, Gakuto Kurata

IBM Tokyo Research Lab, Tokyo, Japan

sashi.novitasari1@ibm.com, fukuda1@jp.ibm.com, gakuto@jp.ibm.com

Abstract

E2E mixed-case (MC) ASR is a more challenging task than unicast (UC) ASR because of the necessity of capitalizing and punctuating the decoded outputs simultaneously. MC models that are simply trained on formatted transcriptions often suffer from various negative impacts, notably a degradation in case-and-punctuation-insensitive performance due to the increased learning complexity. In this paper, we describe novel techniques for training E2E MC ASR models and use them to improve casing-and-punctuation sensitive and insensitive performance. Our approach incorporates knowledge distillation from UC teacher to MC student models not only to improve capitalization and punctuation accuracy but also to maximize phone classification capability in MC ASR. Furthermore, we attempt to integrate voice activity cues into MC ASR to support text formatting tasks. Our method significantly reduces errors by up to 9.2% relative to baseline models that operate at a similar decoding cost.

Index Terms: End-to-end mixed-case ASR, capitalization, punctuation, knowledge distillation, voice activity detection

1. Introduction

End-to-end (E2E) automatic speech recognition (ASR) frameworks such as connectionist temporal classification (CTC) [1], recurrent neural network transducer (RNN-T) [2, 3], and Conformer [4, 5] have recently been used for directly transforming speech audio into transcription text through a single model. Those systems require transcribed speech data in order to train the single all-neural models. Thus, given that data collection covering multiple domains can be expensive, the limitation of having paired speech-text data becomes profound when building systems that cover larger and more diverse acoustic and language spaces. Since the majority of speech corpora for E2E ASR model training are paired only with non-punctuated unicast (UC) text because of high annotation costs, one compromise is to construct E2E ASR model as-is to output only unicast text without punctuation marks at the sacrifice of readability. Unfortunately, E2E ASR frameworks are often not as customizable as hybrid systems [6, 7] that can be modularly customized for the output formats related to casing and punctuation marks. To solve this problem, reformation of unicast decoded outputs by using an external text processor has been investigated in the literature [8–12]. While, in this case, the problems of capitalization/punctuation and phone classification can be split, the two-stage processing will increase the overall system complexity because more modules are involved in the system, which is often not preferable.

On the other hand, E2E mixed-case (MC) ASR directly decodes input speech into capitalized text with punctuation marks

by training the model using speech data paired with truecased-punctuated transcriptions. Recently, studies that explicitly address the capitalization and punctuation tasks for E2E ASR [13–18] have been gaining attention for demands for practical systems. For instance, several studies have showcased frameworks that directly produce unicast and mixed-case texts by training a model using both formatted and unformatted speech transcription data [15, 16, 19]. They demonstrated the possibility of improving MC ASR performance with a limited amount of training data. In another approach, a UC ASR model was extended with additional output layers for the casing and punctuation label classifications, in which the unicast text output is reformatted on the basis of those labels [14, 17, 18]. Large-scale ASR models, such as Whisper [20], have also been receiving attention for their text formatting capability, although they may incur accuracy-cost trade-offs.

Despite the MC ASR framework’s advantage in simplifying the system, it might have lower phone classification accuracy in comparison with UC ASR [13, 21] because of the increased complexity of the model training, a phenomenon that was also observed in our study. Unlike a UC ASR model that models only a single written form of a word, an MC ASR model needs to not only identify words acoustically but also model several graphical forms of an utterance with different casing and punctuation marks; implying a larger search space than that of UC ASR. Capitalization and punctuation tasks are difficult in various aspects because there are no explicit sounds uttered by humans that correspond to them, which causes the learning complexity in the ASR models to increase. In particular, these problems appear in systems designed with the greedy decoding and situations where no external language model is available.

In this paper, we propose training techniques for E2E MC ASR to enhance both casing-and-punctuation sensitive and insensitive word accuracy, of which the latter has been rarely analyzed through a comparison with UC ASR [14, 15, 18] despite the aforementioned necessity. By using the case-and-punctuation-insensitive metric, we examined details of the performance of word classification itself that might be less noticeable in case-and-punctuation-sensitive metrics, such as missing words, because casing or punctuation-related errors often predominate all others. In this paper, we only dealt with constructing MC models with a similar structure as UC models or with minor changes for decoding to maintain the overall system complexity.

We propose two approaches to achieve our goal. First, we construct an E2E MC system with a knowledge distillation (KD) training procedure [22] to transfer discriminative acoustic knowledge represented in a UC model (as a teacher) into an MC model (as a student) and improve the case-and-punctuation insensitive performance in MC ASR. Since the UC model has

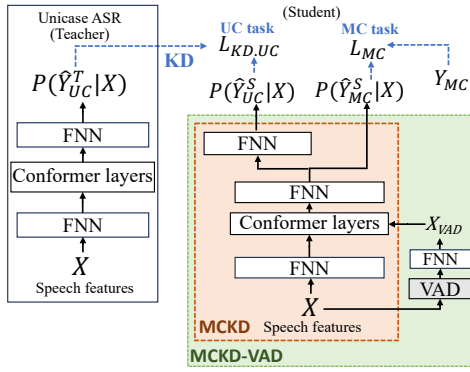


Figure 1: The proposed MCKD ASR (brown box) and MCKD-VAD ASR (green box).

abundant knowledge specific to the unicast transcription task, which is much simpler than the mixed-case task, we hypothesize that it can capture meaningful representations for recognizing words correctly; thus, it could be useful for MC ASR to produce more accurate transcriptions. To the best of our knowledge, there is little prior research that has utilized KD particularly for E2E MC ASR task.

To enhance the capitalization and punctuation accuracy, the second approach extends our method for embedding the E2E MC ASR framework with voice activity cues obtained from a voice activity detection (VAD) system [23–27]. ASR systems are commonly paired with a VAD system that removes the non-speech parts to reduce computational costs while decoding. Since the VAD can distinguish the voiced and non-voiced audio frames by estimating their likelihoods, we hypothesize that feeding this likelihood information into the MC model will also be useful for predicting the punctuation marks and casing. In practice, if the ASR has been paired with a standalone VAD system, the overall proposed system complexity will be similar to that of the standard systems without the VAD embedding. In summary, our contributions are as follows.

1. We utilize KD to construct an MC model from a UC model to improve the case-and-punctuation insensitive performance.
2. We integrate VAD-based information into the MC model to leverage acoustic clues for accurate punctuation and casing.
3. We demonstrate E2E MC ASR that leverages both KD and VAD information in training and show that it performs robustly in inference with or without VAD information.

2. Proposed Modeling Technique

2.1. Baseline fully-supervised multitask MC ASR (MCS)

In prior studies [15, 16], E2E MC ASR systems were designed to directly produce mixed-case text as their main task along with unicast text as an auxiliary task through a fully-supervised multitask training approach. For convenience, we will refer to this framework as the fully-supervised multitask mixed-case (MCS) ASR. Given an acoustic feature sequence $X \in \mathbb{R}^{I \times A}$, an MCS ASR model decodes it into an mixed-case text $Y_{MC} \in \mathbb{R}^{J_{MC}}$ that has punctuation marks and a unicast text $Y_{UC} \in \mathbb{R}^{J_{UC}}$ through separated output layers, where I is the length of the acoustic-features sequence, A is the acoustic feature’s dimension, and J_{MC} and J_{UC} are the lengths of the corresponding texts. Y_{UC} vocabulary consists only of lowercase (or uppercase) characters, while Y_{MC} vocabulary contains lowercase characters, uppercase characters, and punctuation marks. The MCS model is optimized through a combined loss L_{MCS} based on the reference

transcriptions in the corresponding formats:

$$L_{MCS} = \lambda L_{MC} + \alpha L_{UC}. \quad (1)$$

In our experiments, we used the non-autoregressive Conformer ASR framework [4] with a CTC objective [1] as our base framework. Thus, $L_{MC} = -\log P_{CTC}(Y_{MC}|X)$ and $L_{UC} = -\log P_{CTC}(Y_{UC}|X)$ are, respectively, the MC output supervised loss and the UC output supervised loss based on a CTC loss formulation, while λ and α are coefficients for the respective loss terms. On the basis of our preliminary experiments, we decided to attach the auxiliary output layer for the UC task to the top of the main output for MC task, which it also the basis for the proposed frameworks. In inference phase, the auxiliary layer is removed, therefore, MCS ASR computational complexity is similar to that of UC ASR model.

2.2. MC ASR with knowledge distillation (MCKD)

We propose a training framework for an MC ASR system with knowledge distillation (MCKD) from a teacher UC model. In this paper, the UC and MC models are each based on a non-autoregressive Conformer-CTC model. The architecture is the same for both models except the output layer. During training, the proposed MCKD (Fig. 1) employs a multitask structure that is optimized on two tasks simultaneously: 1) punctuated and capitalized MC speech recognition as the main task, and 2) UC speech recognition as the auxiliary task. MCKD ASR’s output layer is extended with an auxiliary output branch, where the KD loss is calculated between this auxiliary UC output and the teacher UC model output and it back-propagates through the entire MCKD network. After the training, the auxiliary layer is removed for inference; therefore, the complexity of the MCKD ASR model remains unchanged from the UC ASR model.

The proposed model training process involves the KD loss and supervised loss. The KD loss ($L_{KD,UC}$) is calculated between the unicast soft outputs from the UC ASR model ($\hat{Y}_{UC}^T$) and the MCKD auxiliary unicast output ($\hat{Y}_{UC}^S$):

$$L_{KD,UC} = \text{KD Loss}(P_{CTC}(\hat{Y}_{UC}^T|X), P_{CTC}(\hat{Y}_{UC}^S|X)). \quad (2)$$

In our preliminary experiments, we explored a mean-square error (MSE) function and the Kullback–Leibler divergence formulation for our KD loss function. We update the model by using the KD loss combined with the supervised ASR loss for MC text (L_{MC}):

$$L_{MCKD} = \lambda L_{MC} + \beta L_{KD,UC}, \quad (3)$$

where λ and β are coefficients for the corresponding loss terms. L_{MC} is calculated in a supervised manner based on the CTC formulation, similarly to the baseline MCS.

2.3. MCKD with voice activity cues (MCKD-VAD)

We further extend the proposed MCKD training framework with the VAD information integration mechanism (MCKD-VAD), as shown in Fig. 1. VAD module is commonly used to classify each signal frame in an audio stream as to its voice activity type (e.g. speech, silence), which it separately works as an external module to ASR system. The VAD output is then utilized to determine the start and end points of audio segments with speech, where the detected segments along with some additional onset and offset frames are to be passed to ASR. Even in a speech segment determined by the VAD model, the likelihoods of the speech frames are much different from each other because of consonants, vowels, lip noise, and so on. Thus, we assume that this information is useful as an auxiliary component for phone classification, casing, and the punctuation task in MC ASR.

The proposed method incorporates a standalone neural VAD model [25] into the ASR model training framework and/or for inference of ASR decoding. VAD integration begins with a voice activity prediction by the VAD model associated with input speech feature sequence X . The output from the VAD model before the softmax is then projected to the same dimension as that of the ASR Conformer layer's, which can be formulated as:

$$X_{\text{VAD}} = \text{FNN}(\text{VAD}(X)), \quad (4)$$

where $X_{\text{VAD}} \in \mathbb{R}^{I \times D}$ is the projected VAD output and D is the dimension of the Conformer layer. X_{VAD} is then integrated into the intermediate Conformer layer inside the ASR model. Integrating additional low-level acoustic features into the intermediate layer is intended to make the model effectively take new information without hindering the original internal log-Mel features and leverage them together in the deeper layers. A similar idea can be also applied to other types of neural network framework, i.e., it is not limited to the Conformer. Suppose that the Conformer network consists of l layers, we add the VAD information X_{VAD} to the $l/2$ -th Conformer layer output ($X_{l/2}$):

$$X_{l/2, \text{final}} = X_{l/2} + \gamma X_{\text{VAD}}, \quad (5)$$

where $X_{l/2, \text{final}}$ is the final output of the intermediate Conformer layer that will be passed to the next Conformer layer, and γ is the coefficient for X_{VAD} . The proposed MCKD-VAD model is optimized with the same objective as the MCKD model (Eq. 3). We freeze the VAD model during the MCKD-VAD training and update only the projection layer for VAD and the model components for ASR.

We examine a MCKD-VAD model that performs robustly with or without integrating the VAD information X_{VAD} into the Conformer during inference. This enables the proposed system to be also used in the same inference setting as the UC ASR system. In our setting, we use $\gamma=1$ for half of the training batches, denoting that VAD information is applied, and $\gamma=0$ for the other half, denoting that the VAD information is not applied, in turns.

3. Experiment Settings

3.1. Data

In our experiments, we consider realistic ASR conditions such as the telephony and meeting conversations, where MC tasks might be more challenging. Our ASR models were trained using in-house English speech-text data that mainly consisted of call-center conversations between multiple speakers, including males and females, recorded under multi-acoustic conditions. The teacher UC model was trained with more than 10K hours of those data, while the student MC model was trained on more than 5K hours of speech data, which was a subset of the UC model training data, with the capitalized and punctuated transcriptions. The VAD model was trained using 2K hours of in-house audio data consisting of spontaneous speech, read speech, and non-speech (noise) audio. All of the audio data were sampled at 8 kHz, from which we extracted log-Mel speech features with a window size of 10 ms and augmented them with delta and double delta coefficients. Every two consecutive frames were stacked followed by alternate frame skipping, which resulted in a 240-dimensional feature vector every 20 ms. The output vocabulary of the UC model was composed of 42 characters that included lowercase characters, digits, apostrophe mark, whitespace mark, and blank mark for CTC. The output vocabulary from the MC model consisted of 97 characters after adding the uppercased characters and the punctuation marks into the UC model vocabulary.

We tested our model on in-house and also publicly available datasets. The in-house test data consisted of 1 hour of real call-center conversations between multiple English speakers, including males and females, recorded under clean conditions (CC-clean). Additionally, we tested our model on 1.5 hours of call-center and meeting recordings that contained the environmental noises (CCM-noisy) and recorded from mixed-accent English speakers, including males and females. We also evaluated our model on Fisher LDC2004S13 [28], which is composed of telephony conversation audio and the corresponding truecased-punctuated transcription. We focused on utterances longer than 60 secs, for which capitalizations and punctuations would be practically more meaningful in the corresponding transcriptions.

3.2. Model configuration

The UC and MC models in our experiments had similar Conformer-CTC architecture. The encoder was composed of an FNN with 768 cells and ten Conformer blocks. Each block had two FNN modules in which the multi-head self-attention module and the convolution module were sandwiched. The multi-head self-attention module had $d^{\text{head}}=8$ and $d^{\text{att}}=96$ for the number of heads and the dimensions of the attention module, and the convolution kernel size was 31. The last Conformer block was followed by an FNN output layer. The output layer size was 42 for the UC model and 97 for the MC model. In the baseline MCS and proposed model, the MC output layer was followed by an auxiliary output layer for the UC task. All the models had roughly 155M trainable parameters. We used a learning rate of $5e-5$ and a batch size of 32 for the ASR training. The baseline MC ASR and the proposed MCKD ASR model parameters were initialized from an UC ASR model, which it was also used as the teacher for the proposed MCKD and MCKD-VAD ASR. The proposed MCKD-VAD model parameters were initialized from the MCKD model parameters. Additionally, we applied several regularization techniques such as sequence noise injection [29] and SpecAugment [30] while training both the baseline and proposed models. In MCKD-VAD training, SpecAugment was applied to the teacher only for the use case in noisy situations. The baseline MCS model used $\alpha=0.2$, while MCKD ASR used $\beta=0.2$ and MCKD-VAD used $\beta=0.4$, our best setting. All the models were fixed to $\lambda=1$. For the MCKD-VAD model, we used a speech frame-level VAD [25] that had four CNN layers and classified the input frames into "speech," "non-speech," and "music" classes. The VAD output was processed through three FNN layers before it was integrated into the ASR. No external language model was used in any ASR inference.

3.3. Evaluation metric

Our models were evaluated on the basis of the word error rate (WER) in several text formats [31]. We denote WER_{cp} as the case-and-punctuation-sensitive WER, where the ASR output was evaluated against the truecased-punctuated reference text. Here, the same word in the reference and ASR output but with different casing or punctuation was regarded as a substitution error. WER_{uncp} denotes a case-and-punctuation-insensitive WER, where the reference and ASR output text were all lowercased and the punctuation marks were removed. We also calculated the case-sensitive WER (WER_{c}) by removing the punctuation marks from the reference and ASR output while keeping the casings. To evaluate the punctuation, we use a WER based on punctuation (WER_{p}) that was calculated by replacing all of the alphanumeric words in the reference and ASR text with a uniform token while keeping the punctuation unchanged. For

Table 1: Performance (%) of the baselines and the proposed MCKD on CC-clean set. (“ L_{UC} ”: supervised loss on UC task, “ L_{MC} ”: supervised loss on MC task, “ $L_{KD,UC}$ ”: KD loss on UC task).

Model (training loss)	WER _{cp}	WER _{uncp}	WER _p	WER _c
Baseline ASR				
UC (L_{UC})	45.8	11.8	27.4	28.1
Single-task MC (L_{MC})	29.0	13.1	17.7	18.8
Multitask MC (MCS) ($L_{MC}+L_{UC}$)	28.7	13.1	17.2	18.7
Proposed ASR				
MCKD+MCS ($L_{MC}+L_{KD,UC}+L_{UC}$)	28.0	12.5	17.0	18.2
MCKD ($L_{MC}+L_{KD,UC}$)	27.7	12.0	17.2	18.0

Table 2: Performance (%) of the proposed MCKD-VAD on CC-clean set. Inference with the VAD feature integration was performed by fusing the VAD features into Conformer.

Model	Inference with VAD feature integration	WER _{cp}	WER _{uncp}	WER _p	WER _c
Baseline ASR					
UC	No	45.8	11.8	27.4	28.1
MCS	No	28.7	13.1	17.2	18.7
Proposed ASR (Training loss: $L_{MC}+L_{KD,UC}$)					
MCKD-VAD	No	27.8	12.3	16.6	17.9
MCKD-VAD	Yes	27.1	11.9	16.6	17.4

example, a text ‘Hi, how are you?’ was converted into ‘W, W W W?’ for the WER_p computation.

4. Results and Discussion

4.1. Baseline ASR

Our baselines were the UC ASR model teacher that taught the proposed models, the single-task MC ASR model that optimized only based on MC texts supervisedly, and the fully-supervised multitask MC (MCS) ASR model optimized based on Eq. 1. As seen in Table 1, the single-task MC and MCS had better WER_{cp}, WER_c and WER_p than those of UC ASR, but their WER_{uncp} were degraded, which implies they produced more incorrect words regardless the format. Our observations indicated that the deteriorated WER_{uncp} was mainly caused by the increased number of substitution and deletion errors, affecting both the originally plain words and the cased or punctuated words. Since the learning complexity of MC-based ASR increased when learning multiple variants of text form associated with phones, the MC-based models were more difficult to converge in terms of the case-and-punctuation-insensitive aspect compared with the UC ASR model. For the experiments reported in the next subsection, we focus on MCS as our best MC ASR baseline.

4.2. Proposed MCKD ASR

Table 1 shows the performance of the proposed MCKD model on the clean call center (CC-clean) audio. Initially, we experimented on an “MCKD+MCS” model that applied both the supervised and KD losses on the auxiliary UC task optimization. Although we observed an improvement over the baseline MCS model, better results were produced when using only the KD loss for the UC task, as the learning objective was simpler.

Our best MCKD model reduced WER_{cp} from the 28.7% of MCS model to 27.7% while it also improved WER_{uncp} by 8.4% relative to that of the baseline MCS model and achieved a comparable WER_{uncp} to that of the UC model. These results show that the auxiliary UC task optimization with KD was effective, because it allowed the student network to learn the word classification knowledge that had been correlated with the acoustic information through the teacher network, which it might be more beneficial for the convergence of the MC model.

Table 3: Performance (%) of the proposed ASR on CCM-noisy set.

Model	Inference with VAD feature integration	WER _{cp}	WER _{uncp}	WER _p	WER _c
Baseline ASR					
UC	No	48.4	25.0	21.7	36.9
MCS	No	36.8	26.0	16.0	30.0
Proposed ASR					
MCKD	No	36.5	25.7	16.0	29.7
MCKD-VAD	No	35.9	25.4	15.9	29.4
MCKD-VAD	Yes	35.5	24.5	15.5	28.9

Table 4: Performance (%) of the proposed ASR on Fisher set.

Model	Inference with VAD feature integration	WER _{cp}	WER _{uncp}	WER _p	WER _c
Baseline ASR					
UC	No	31.8	10.8	18.0	19.3
MCS	No	23.2	8.9	15.4	11.4
Proposed ASR					
MCKD	No	22.9	8.9	15.1	11.3
MCKD-VAD	No	22.6	8.7	14.8	11.2
MCKD-VAD	Yes	22.5	8.7	14.8	11.2

4.3. Proposed MCKD-VAD ASR

We carried out the MCKD-VAD ASR inference with and without VAD information integration. The inference with VAD information integration followed the mechanism described in Section 2.3, while the inference without VAD integration had the same mechanism as those of UC, MCS, and MCKD ASR.

As shown in Table 2, on CC-clean, the proposed MCKD-VAD model performed robustly in terms of inferences with and without the VAD integration; it outperformed the baseline MCS and had a similar WER_{cp} and WER_{uncp} to the MCKD ASR when VAD information was not used for inference. Higher gain was observed when VAD features were used during ASR inference, where a relative reduction in WER_{cp} and WER_{uncp} of 5.6% and 9.2 % over the baseline MCS model, respectively, was achieved. Compared with the MCKD ASR model, MCKD-VAD ASR model’s gain was more noticeable in terms of WER_p. We found that punctuation marks such as commas and periods could be associated with voice activity, such as short pauses, in the spoken utterances and that this led to a better punctuation recognition and in turn better casings. With the proposed regulated MCKD-VAD model training mechanism, our model maintained that capability even in the VAD information-less inference.

We additionally experimented on the CCM-noisy (Table 3) and the Fisher test sets (Table 4). The MCKD ASR and MCKD-VAD ASR models outperformed the baselines. Moreover, the MCKD-VAD ASR model, which used the VAD features for inference, outperformed the UC ASR WER_{uncp} on CCM-noisy. This indicates that VAD information was particularly helpful in acoustically more challenging situations, as the likelihood for voiced/non-voiced speech parts might prompt the ASR system’s attention to process those parts more carefully in noises.

5. Conclusion

We proposed a model training algorithm for E2E MC ASR with KD and a VAD cues integration mechanism to improve ASR’s case-and-punctuation sensitive and insensitive performance. Our experimental results showed that the proposed KD from a UC model to a MC model improved the model’s case-and-punctuation insensitive performance, while the VAD information resulted in a case-and-punctuation sensitive performance gain. As for future work, we are interested in exploring other teacher structures for KD and different additional information for the MC network.

6. References

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